

# ABDUL AZIZ

Chittagong-4202, Bangladesh

 [abdul.aziz.cu@gmail.com](mailto:abdul.aziz.cu@gmail.com)  [Google Scholar](#)  [Website](#)  [Faculty Profile](#)

## RESEARCH INTERESTS

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Multimodal AI, Cross-cultural Context Modeling, Multilingual and Low-resource NLP, Vision-Language Modeling, AI for Social Good, LLMs Reasoning and Applications, and Natural Language Processing.

## EDUCATION

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B.Sc. (Engg.) in Computer Science and Engineering (CSE) *Jan 2018 - Oct 2023*

[University of Chittagong](#)

Chittagong, Bangladesh

CGPA: 3.64/4.0 (**Last 3 years: 3.81/4.0; Last 60 credits: 3.88/4.0**; Class rank: 8/61)

Medium of Instruction: English

Undergraduate Thesis: Multimodal Human Desire Understanding on Social Media Data

## RESEARCH EXPERIENCE

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### Research Assistant

*Apr 2020 - Present*

- At [CSECU-DSG Lab](#) with [Dr. Abu Nowshed Chy](#), University of Chittagong, Chittagong, Bangladesh
  - Secured **first place** at **CLEF-2023**, **EMNLP-2022**, and **FIRE-2020** Conference's research track

### Research Collaboration

*Jan 2021 - Present*

- At [KDE Lab](#) with [Prof. Dr. Masaki Aono](#), Toyohashi University of Technology Toyohashi, Japan
  - Publications: J2
- With [Prof. Dr. Muhammad Ashad Kabir](#), Charles Sturt University Estella NSW, Australia
  - Publications: J1
- With [Dr. Md Zia Ullah](#), Edinburgh Napier University Edinburgh, UK
  - Publications: UR1, UR2, UR3, J2, C1

## TEACHING EXPERIENCE

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### Lecturer

*July 2025 - Present*

### Assistant Lecturer

*February 2025 - July 2025*

### Adjunct Lecturer

*July 2024 - January 2024*

Dept. of CSE, [International Islamic University Chittagong \(IIUC\)](#)

Chittagong, Bangladesh

([Experience Certificate](#))

## PUBLICATIONS

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### Under Review Papers (UR):

*\* implies equal contributions*

UR1. [Abdul Aziz](#), Md Sadman Hafiz, Md Monir Hossain, Abu Nowshed Chy, and Md Zia Ullah, *Beyond Categorization: A Multimodal Multitask Framework for Fine-Grained Disaster Severity Estimation in a Low-Resource Language*, **ACM WSDM Conference**, 2027.

UR2. [Abdul Aziz](#), Md. Akram Hossain, Abu Nowshed Chy, and, Md Zia Ullah, *Leveraging Multitask Contrastive Learning for Debiasing Multimodal Complaint Detection*, **Elsevier Information Fusion Journal**.

UR3. Abu Nowshed Chy, [Abdul Aziz](#), and Md Zia Ullah, *Multi-Task Contrastive Learning for Complaint Detection Leveraging Sentiment, Emotion, and Sarcasm-Aware Signals*, **IEEE Transactions on Computational Social Systems Journal**.

## Journal Papers (J):

\* implies equal contributions

- J1. **Abdul Aziz**, Nihad Karim Chowdhury, Muhammad Ashad Kabir, Abu Nowshed Chy, and Md. Jawad Siddique, *MMTF-DES: A Fusion of Multimodal Transformer Models for Desire, Emotion, and Sentiment Analysis of Social Media Data*, **Elsevier Neurocomputing Journal**, Vol. 623, 2025. ([URL](#))
- J2. **Abdul Aziz\***, Md. Akram Hossain\*, Abu Nowshed Chy, Md Zia Ullah, and Masaki Aono, *Leveraging Contextual Representations with BiLSTM-based Regressor for Lexical Complexity Prediction*, **Elsevier Natural Language Processing Journal**, Vol. 05, 2023. ([PDF](#))

## Selected Conference and Workshop Papers (C) :

\* implies equal contributions

- C1. Md. Akram Hossain, **Abdul Aziz**, Muhammad Anwarul Azim, Abu Nowshed Chy, Md Zia Ullah, and Mohammad Khairul Islam, *BiGCAT: A Graph-Based Representation Learning Model with LLM Embeddings for Named Entity Recognition*, **RANLP 2025**, Varna, Bulgaria (Long Paper). ([PDF](#))
- C2. Nabila Ayman, Md. Akram Hossain\*, **Abdul Aziz\***, Rokan Uddin Farouqi, and Abu Nowshed Chy, *BengaliLCP: A Dataset for Lexical Complexity Prediction in the Bengali Texts*, **LREC-COLING 2024**, Torino, Italia (Long Paper). ([PDF](#))
- C3. **Abdul Aziz**, Md. Akram Hossain, and Abu Nowshed Chy, *CSECU-DSG at CheckThat! 2023: Transformer-based Fusion Approach for Multimodal and Multigenre Check-Worthiness*, **CLEF 2023**, Thessaloniki, Greece. ([PDF](#))
- C4. **Abdul Aziz**, Md. Akram Hossain, and Abu Nowshed Chy, *CSECU-DSG@ Multimodal Hate Speech Event Detection 2023: Transformer-based Multimodal Hierarchical Fusion Model For Multimodal Hate Speech Detection*, **CASE@RANLP 2023**, Varna, Bulgaria. ([PDF](#))
- C5. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *CSECU-DSG at SemEval-2023 Task 4: Fine-tuning DeBERTa Transformer Model with Cross-fold Training and Multi-sample Dropout for Human Values Identification*, **SemEval@ACL 2023**, Toronto, Canada. ([PDF](#))
- C6. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *CSECU-DSG @ Causal News Corpus 2022: Fusion of RoBERTa Transformers Variants for Causal Event Classification*, **CASE@EMNLP 2022**, Abu Dhabi, United Arab Emirates. ([PDF](#))
- C7. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *Enhancing the DeBERTa Transformers Model for Classifying Sentences from Biomedical Abstracts*, **ALTA 2022**, Adelaide, Australia. ([PDF](#))
- C8. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *CSECU-DSG at SemEval-2022 Task 3: Investigating the Taxonomic Relationship Between Two Arguments using Fusion of Multilingual Transformer Models*, **SemEval@NAACL 2022**, Washington, USA. ([PDF](#))
- C9. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *CSECU-DSG at SemEval-2021 Task 1: Fusion of Transformer Models for Lexical Complexity Prediction*, **SemEval@ACL-IJCNLP 2021**, Bangkok, Thailand. ([PDF](#))
- C10. **Abdul Aziz\***, Md. Akram Hossain\*, and Abu Nowshed Chy, *Feature Fusion with Hand-crafted and Transfer Learning Embeddings for Cause-Effect Relation Extraction*, **CEREX@FIRE 2020**, Hyderabad, India. ([PDF](#))

## ACHIEVEMENTS

**1st Place (Winning Team) in CheckThat!@CLEF 2023 Subtask 1A Arabic** Aug 2023  
Task and Paper: Check-Worthiness in Multimodal and Multigenre Content ([Task Website](#); [Paper PDF](#))

**1st Place (Winning Team) in CASE@EMNLP 2022 Subtask 1** Dec, 2022  
Task and Paper: Event Causality Identification ([Task Website](#); [Paper PDF](#))

**1st Place (Winning Team) in CEREX@FIRE 2020 Task A***Dec, 2020*

Task and Paper: Automatic Cause-effect Relation Extraction from Text ([Task Website](#); [Paper PDF](#))

**2nd Place in ALTA 2022 Shared Task***Dec, 2022*

Task and Paper: Abstracts Classification for Evidence-Based Medicine ([Task Website](#); [Paper PDF](#))

**4th Place in PreTENS-SemEval@NAACL 2022 Subtask 1***Jul, 2022*

Task and Paper: Presupposed Taxonomies (PreTENS) ([Task Website](#); [Paper PDF](#))

**EXTRA RESEARCH ACTIVITIES AND SKILLS**

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**Reviewer:** ACL '25, CLEF '23, SemEval '22;

**Conference Student Volunteer:** ACL '23, SIGIR '21

**Programming Languages:** Python, C, C++, JAVA;

**Human Languages:** Bengali (*Native*), English (*Proficient in all aspects*)

**ML Libraries:** PyTorch (*implement most of the projects using it*), TensorFlow, Huggingface

**Coursera Online Certifications:** ([Certification's PDF](#))

-Natural Language Processing Specialization

-Deep Learning Specialization

-DeepLearning.AI TensorFlow Developer Specialization.

**REFEREES**

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Dr. Abu Nowshed Chy

Assistant Professor

Computer Science & Engineering,

University of Chittagong, Chittagong-4331, Bangladesh

Phone: +8801764207358

Email: nowshed@cu.ac.bd

Dr. Md Zia Ullah

Lecturer (Assistant Professor),

School of Computing, Engineering

and the Built Environment,

Edinburgh Napier University, Edinburgh, UK

Phone: +8801738857061

Email: M.Ullah@napier.ac.uk